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CECS 327 - Project 2

Task 1: Anycast with Docker (TCP)

Example outputs:
`docker-compose up –build`

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

 => => unpacking to docker.io/library/task1-anycast-with-docker-server2:latest                                    0.2s
 => [server1] exporting to image                                                                                  0.9s
 => => exporting layers                                                                                           0.6s
 => => exporting manifest sha256:832439cdb369140673d3e787cf2629ea7a73352cc177c8c61897eefffa8b5895                 0.0s
 => => exporting config sha256:c6438271f4dd19ec2da2916fce79dec1f639c161105cda36f8ce518932e3511b                   0.0s
 => => exporting attestation manifest sha256:c9fac0fb780cd213c3f4b4d6deac34e876b21e6700620235898 97ab0cfcb7559     0.0s
 => => exporting manifest list sha256:7e617970f94ace04d2d5c22a8c8421454bc893935b21881816a9206336f9684d            0.0s
 => => naming to docker.io/library/task1-anycast-with-docker-server1:latest                                       0.0s
 => => unpacking to docker.io/library/task1-anycast-with-docker-server1:latest                                    0.2s
 => [client] resolving provenance for metadata file                                                               0.0s
 => [server1] resolving provenance for metadata file                                                              0.0s
 => [server3] resolving provenance for metadata file                                                              0.0s
 => [server2] resolving provenance for metadata file                                                              0.0s
[+] up 9/9
 ✔ Image task1-anycast-with-docker-server3       Built                                                            5.3s
 ✔ Image task1-anycast-with-docker-client        Built                                                            5.3s
 ✔ Image task1-anycast-with-docker-server1       Built                                                            5.3s
 ✔ Image task1-anycast-with-docker-server2       Built                                                            5.3s
 ✔ Network task1-anycast-with-docker_default     Created                                                          0.0s
 ✔ Container task1-anycast-with-docker-server2-1 Created                                                          0.1s
 ✔ Container task1-anycast-with-docker-server1-1 Created                                                          0.1s
 ✔ Container task1-anycast-with-docker-server3-1 Created                                                          0.1s
 ✔ Container task1-anycast-with-docker-client-1  Created                                                          0.1s
Attaching to client-1, server1-1, server2-1, server3-1
server2-1   | Server ready on port 5000
server3-1   | Server ready on port 5000
server1-1   | Server ready on port 5000
server1-1   | Accepted connection from ('172.19.0.5', 55644)
server1-1   | Sent: Hello from server1 !!!


server3-1   | Accepted connection from ('172.19.0.5', 59368)
server2-1   | Accepted connection from ('172.19.0.5', 54068)



server3-1   | Sent: Hello from server3 !!!
client-1    | Received: Hello from server1 !!!
server2-1   | Sent: Hello from server2 !!!


client-1    | Received: Hello from server2 !!!
client-1    | Received: Hello from server3 !!!
client-1 exited with code 0
server1-1   | Accepted connection from ('172.19.0.5', 60520)
server1-1   | Sent: Hello from server1 !!!

server2-1   | Accepted connection from ('172.19.0.5', 43044)
server3-1   | Accepted connection from ('172.19.0.5', 36540)ed: Hello from server1 !!!
task1-anycast-with-docker-client-run-395f4c01a62f   | Received: Hello from server2 !!!


server2-1                                           | Sent: Hello from server2 !!!
server3-1                                           | Sent: Hello from server3 !!!
task1-anycast-with-docker-client-run-395f4c01a62f exited with code 0

v View in Docker Desktop   o View Config   w Enable Watch   d Detach
```

`tcpdump` logs

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

❯ docker exec -it task1-anycast-with-docker-server1-1 tcpdump -i eth0 port 5000
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
20:16:26.973712 IP task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520 > d1ab09d5bb09.5000: Flags [S], seq 1382425849, win 64240, options [mss 1460,sa
ckOK,TS val 1609978971 ecr 0,nop,wscale 7], length 0
20:16:26.973732 IP d1ab09d5bb09.5000 > task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520: Flags [S.], seq 4044959396, ack 1382425850, win 65160, opt
ions [mss 1460,sackOK,TS val 3767662359 ecr 1609978971,nop,wscale 7], length 0
20:16:26.973746 IP task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520 > d1ab09d5bb09.5000: Flags [.], ack 1, win 502, options [nop,nop,TS val 1609978
971 ecr 3767662359], length 0
20:16:26.973828 IP d1ab09d5bb09.5000 > task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520: Flags [P.], seq 1:24, ack 1, win 510, options [nop,nop,TS
val 3767662359 ecr 1609978971], length 23
20:16:26.973840 IP task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520 > d1ab09d5bb09.5000: Flags [.], ack 24, win 502, options [nop,nop,TS val 160997
8971 ecr 3767662359], length 0
20:16:26.973849 IP d1ab09d5bb09.5000 > task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520: Flags [F.], seq 24, ack 1, win 510, options [nop,nop,TS va
l 3767662359 ecr 1609978971], length 0
20:16:26.973882 IP task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520 > d1ab09d5bb09.5000: Flags [F.], seq 1, ack 25, win 502, options [nop,nop,TS va
l 1609978971 ecr 3767662359], length 0
20:16:26.973885 IP d1ab09d5bb09.5000 > task1-anycast-with-docker-client-run-395f4c01a62f.task1-anycast-with-docker_default.60520: Flags [.], ack 2, win 510, options [nop,nop,TS val 3767662
359 ecr 1609978971], length 0
```

`docker-compose run client`

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

❯ ls
README.md                    proj1-docker-containers        proj2-distributed-communication
❯ cd proj2
cd: no such file or directory: proj2
❯ cd proj2-distributed-communication
❯ ls
task1-anycast-with-docker
❯ cd task1-anycast-with-docker
❯ ls
Dockerfile              client.py              docker-compose.yml      server.py
❯ docker-compose run client
[+]  3/3t 3/33
 ✔ Container task1-anycast-with-docker-server2-1 Running                                    0.0s
 ✔ Container task1-anycast-with-docker-server1-1 Running                                    0.0s
 ✔ Container task1-anycast-with-docker-server3-1 Running                                    0.0s
Container task1-anycast-with-docker-client-run-395f4c01a62f Creating
Container task1-anycast-with-docker-client-run-395f4c01a62f Created
Received: Hello from server1 !!!
Received: Hello from server2 !!!
Received: Hello from server3 !!!

~/s/e/u/sp26/32/327_Docker/proj2/task1-anycast-with-docker on main !2 ····· at 13:16:27
❯
```

Task 2: Multicast with UDP

Example outputs:

`docker-compose up –build`

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

❯ clear
❯ docker-compose up
[+] up 5/5
 ✔ Network task2-multicast-with-udp_default       Created                                   0.0s
 ✔ Container task2-multicast-with-udp-sender2-1   Created                                   0.0s
 ✔ Container task2-multicast-with-udp-sender1-1   Created                                   0.0s
 ✔ Container task2-multicast-with-udp-receiver1-1 Created                                   0.0s
 ✔ Container task2-multicast-with-udp-receiver2-1 Created                                   0.0s
Attaching to receiver1-1, receiver2-1, sender1-1, sender2-1
sender1-1   | Sender started — broadcasting as sensor: temp
sender1-1   | Sent: {"sensor": "temp", "value": 29.8}
sender1-1   | Sent binary data: b'\x7f\x8b\x1c\xb4\n\x03\xe2\xd9'
sender2-1   | Sender started — broadcasting as sensor: humidity
sender2-1   | Sent: {"sensor": "humidity", "value": 25.5}
sender2-1   | Sent binary data: b'\x84%\xf3\xd1T\x92\x86\x91'
receiver2-1 | Joined multicast group
receiver1-1 | Joined multicast group

receiver1-1 | Received: {'sensor': 'temp', 'value': 27.6} from ('172.19.0.2', 60319)

receiver1-1 | Received binary data from ('172.19.0.2', 60319): b'\x06d\xc6\x9d\x81X\xac('
receiver2-1 | Received binary data from ('172.19.0.2', 60319): b'\x06d\xc6\x9d\x81X\xac('
receiver2-1 | Received: {'sensor': 'humidity', 'value': 20.7} from ('172.19.0.3', 59796)

receiver1-1 | Received: {'sensor': 'humidity', 'value': 20.7} from ('172.19.0.3', 59796)
sender2-1   | Sent: {"sensor": "humidity", "value": 20.7}
receiver2-1 | Received binary data from ('172.19.0.3', 59796): b']UZ\xaa\xa8\x08oY'

sender2-1   | Sent binary data: b']UZ\xaa\xa8\x08oY'atch    d Detach

receiver1-1 | Received binary data from ('172.19.0.3', 59796): b']UZ\xaa\xa8\x08oY'

sender1-1   | Sent: {"sensor": "temp", "value": 27.7}
receiver2-1 | Received: {'sensor': 'temp', 'value': 27.7} from ('172.19.0.2', 60319)
receiver1-1 | Received: {'sensor': 'temp', 'value': 27.7} from ('172.19.0.2', 60319)

receiver1-1 | Received binary data from ('172.19.0.2', 60319): b'c\xee\x8b\x88\x9f\xebtS'
receiver2-1 | Received binary data from ('172.19.0.2', 60319): b'c\xee\x8b\x88\x9f\xebtS'

receiver1-1 | Received: {'sensor': 'humidity', 'value': 26.7} from ('172.19.0.3', 59796)
sender2-1   | Sent: {"sensor": "humidity", "value": 26.7}

Ln 11, Col 1   Spaces: 4   UTF-8   LF   { } Python   3.13.7   Python 3.14.3 (homebrew)
```

`tcpdump` logs

```
❯ docker exec -it task2-multicast-with-udp-receiver1-1 tcpdump -i eth0 udp port 5007
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
23:28:07.073599 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:07.073599 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:07.073677 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:07.073771 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:08.077126 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:08.077278 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:08.077815 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:08.077908 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:09.082122 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:09.082122 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:09.082172 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:09.082290 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:10.086629 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:10.086772 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:10.087213 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:10.087299 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:11.088071 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:11.088086 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:11.088164 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:11.088174 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:12.092209 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:12.092361 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:12.092505 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:12.092686 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:13.094945 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:13.094946 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:13.095082 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:13.095082 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:14.096833 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:14.096833 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:14.096891 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:14.096891 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:15.103269 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:15.103268 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:15.103433 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:15.103439 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
23:28:16.106229 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 37
23:28:16.106229 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 33
23:28:16.106560 IP task2-multicast-with-udp-sender1-1.task2-multicast-with-udp_default.53545 > 224.1.1.1.5007: UDP, length 8
23:28:16.106560 IP task2-multicast-with-udp-sender2-1.task2-multicast-with-udp_default.45188 > 224.1.1.1.5007: UDP, length 8
```

Docker logs:

`docker logs task2-multicast-with-udp-receiver1-1`

```
❯ docker logs task2-multicast-with-udp-receiver1-1
Joined multicast group
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 29.2}
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 27.1}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\xc4{l\xf9\xcf\x01\x02\xdf'
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\x07B\xd7\xbe\x19\x96%\xd7'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 23.2}
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 25.9}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'"\xfdB\xb9\xd8u\xee\xf8'
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\xa87\x9b$\xaes\x83\x16'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 25.3}
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\x9fy-[G\xcc\x18\xaf'
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 25.3}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\xe2\xaf\x93\x91\xa54\xf5\xd0'
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 28.1}
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 23.9}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\x16\x03\x8fG\x9b\x14\x8f\x00'
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\x8d\x8c!\xae}\xca\xbf\x9c'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 28.1}
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 22.5}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\xfeI\x01\xb0)\xfb\xed7'
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'w\xd7\xb1\xc9\xa8\xbe8\xb0'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 25.1}
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 27.1}
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\xd3\x0e\xe9\x08\xb8\xd2\x8b|'
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b'\x1a\xc9n\x1b\xe32K1'
Received: Multicast message from ('172.19.0.3', 44968)
Received: {'sensor': 'humidity', 'value': 23.5}
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 20.2}
Received: Multicast message from ('172.19.0.3', 44968)
Received binary data from ('172.19.0.3', 44968): b";\xd5\x9b\x14\x0fn'c"
Received: Multicast message from ('172.19.0.2', 53790)
Received binary data from ('172.19.0.2', 53790): b'\x9ed|Qf\\]\xe3'
Received: Multicast message from ('172.19.0.2', 53790)
Received: {'sensor': 'temp', 'value': 24.1}
Received: Multicast message from ('172.19.0.2', 53790)
```

`docker logs task2-multicast-with-udp-sender2-1`

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

❯ docker logs task2-multicast-with-udp-sender2-1
Sender started — broadcasting as sensor: humidity
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 26.9}
Sent binary data: b'\x19c\x12\x97,\x02\x19\xda'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 29.2}
Sent binary data: b'\xc4{l\xf9\xcf\x01\x02\xdf'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.9}
Sent binary data: b'"\xfdB\xb9\xd8u\xee\xf8'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.3}
Sent binary data: b'\xe2\xaf\x93\x91\xa54\xf5\xd0'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 28.1}
Sent binary data: b'\x16\x03\x8fG\x9b\x14\x8f\x00'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 22.5}
Sent binary data: b'\xfeI\x01\xb0)\xfb\xed7'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 27.1}
Sent binary data: b'\x1a\xc9n\x1b\xe32K1'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 23.5}
Sent binary data: b";\xd5\x9b\x14\x0fn'c"
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 22.4}
Sent binary data: b'L\x8e\x04\x1a\t\xf7\xc2r'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.3}
Sent binary data: b'\x97u\x9f\x97\x17\xfb\xc9@'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 23.3}
Sent binary data: b'p\xc2\xe9\xd0+\xb8\x0f\x19'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 28.4}
Sent binary data: b'\xdb=\xad\xac\xda\xe6\xf9\x8a'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 20.2}
Sent binary data: b'E\xba\xe7 \x1d\x85\xb4\xb0'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 22.9}
Sent binary data: b'/|\x1c\xbf<\xd2\xaf\x87'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 22.2}
Sent binary data: b'\x82y\xf6V\xcc\xce\xf8\xff'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 27.0}
Sent binary data: b"F\x8c\xba'-\x1d:\x96"
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 21.3}
Sent binary data: b'#9\xfb\x9d\x82\x10\x9bh'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.3}
Sent binary data: b'\x18\xf2\xc6r\t\x0e\x8co'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 23.0}
Sent binary data: b'\xc6\r,h\x1dx\x19\x9e'
Sent: Multicast message
Sent: {"sensor": "humidity", "value": 25.1}

zsh task2-...
zsh

Ln 26, Col 5   Spaces: 4   UTF-8   LF   { } Python   3.13.7   Python 3.14.3 (homebrew)
```